

AI Fraud Detection In Crypto-Currency Using Machine learning.

Harsh Wadhwa

*Department of Computer Science and Engineering, Chitkara University, Punjab, India
harsh382.be22@chitkara.edu.in*

Abstract—The rapid growth of cryptocurrency has transformed the global financial ecosystem by enabling decentralized, borderless, and secure digital transactions. However, this rapid adoption has also attracted a significant increase in fraudulent activities such as phishing attacks, Ponzi schemes, fake investment platforms, transaction laundering, and suspicious wallet behaviors. Due to the anonymous and distributed nature of blockchain networks, detecting such fraudulent activities using traditional rule-based financial monitoring systems has become increasingly difficult.

This research explores the use of machine learning techniques to develop an intelligent fraud detection framework for cryptocurrency transactions. By analyzing historical transaction data, wallet interaction patterns, and behavioral features, the proposed system aims to identify anomalies and classify potentially fraudulent activities with improved accuracy. Various supervised learning algorithms including Logistic Regression[5], Decision Tree[7], and Random Forest are implemented and evaluated to determine their effectiveness in handling large-scale and imbalanced cryptocurrency datasets.

Keywords—Cryptocurrency, Fraud Detection, Machine Learning, Blockchain, Anomaly Detection, Artificial Intelligence, Transaction Analysis.

I. INTRODUCTION

Over the past decade, cryptocurrency has emerged as one of the most transformative innovations in the financial and technological landscape [1]. By leveraging blockchain technology, cryptocurrencies enable decentralized and transparent digital transactions without the need for traditional financial intermediaries such as banks or payment institutions. However, this rapid growth has also introduced significant security challenges, including fraud, scams, and money laundering [2].

Unlike conventional financial systems that rely on centralized monitoring and identity verification mechanisms, cryptocurrency networks often operate with a high degree of anonymity. While this anonymity protects user privacy, it also makes it easier for malicious actors to conduct fraudulent transactions, laundering operations, phishing scams, Ponzi investment schemes, and wallet theft. As transaction volumes continue to increase across blockchain platforms, detecting suspicious activities using traditional rule-based or manual monitoring approaches has become increasingly ineffective and time-consuming [5].

Machine learning techniques such as classification and anomaly detection have been widely applied in fraud detection systems to identify suspicious transaction patterns [5]. GRU, and subsequently transformer-based models such as BERT and its successors — has substantially expanded the representational capacity available for mental health NLP. These architectures can model long-range dependencies, contextual word meanings, and discourse-level coherence in ways that lexicon-based approaches cannot, and their ability to be fine-tuned on domain-specific data allows pre-existing linguistic knowledge to be adapted to the specific characteristics of mental health text [4].

In recent years, machine learning has gained significant attention as a powerful tool for identifying hidden patterns and anomalies in large datasets. By analyzing historical transaction data, behavioural patterns of wallets, and

network relationships between users, machine learning models can assist in automatically classifying transactions

as legitimate or potentially fraudulent. This data-driven approach offers a scalable and adaptive solution compared to static fraud detection systems that struggle to keep pace with evolving cybercrime techniques in decentralized environments.

The study investigates how different supervised learning algorithms can be utilized to improve fraud detection accuracy while handling challenges such as imbalanced datasets, feature complexity, and dynamic transaction behavior. By integrating artificial intelligence with blockchain analytics, the proposed framework aims to contribute toward enhancing financial security and strengthening trust in cryptocurrency ecosystems.

Furthermore, the findings of this research may support the development of intelligent monitoring systems for cryptocurrency exchanges,

II. RELATED WORK

A. Classical Machine Learning Approaches

Early research on cryptocurrency fraud detection primarily focused on traditional machine learning techniques applied to structured transaction datasets. Researchers analysed features such as transaction frequency, wallet connectivity, transaction amount distribution, and temporal patterns to identify suspicious behaviours. Studies using datasets such as the Elliptic Bitcoin Transaction dataset demonstrated that classification algorithms like Logistic Regression[5], Decision Trees, and Support Vector Machines could effectively distinguish between legitimate and fraudulent transactions.

Among these approaches, ensemble learning methods such as Random Forest gained popularity due to their ability to handle high-dimensional data and improve predictive performance. These models were particularly useful in addressing challenges related to class imbalance, where fraudulent transactions represent only a small fraction of the total transaction volume. While classical machine learning methods provided a strong baseline for fraud detection, their performance often depended heavily on manual feature engineering and domain expertise.

B. Graph-Based and Network Analysis Approaches

With the recognition that cryptocurrency transactions form complex network structures, researchers began exploring graph-based analytical techniques. In blockchain ecosystems, wallets and transactions can be represented as nodes and edges in a transaction graph. By analyzing network properties such as centrality measures, clustering coefficients, and community detection patterns, studies have identified hidden relationships between fraudulent entities.

Graph analytics methods have shown effectiveness in detecting coordinated fraud schemes such as laundering networks and Ponzi structures. Some research introduced graph embeddings and network representation learning to improve fraud classification accuracy. However, these approaches often require significant computational resources and may face scalability challenges when applied to real-time transaction streams.

C. Transformer and Advanced AI Approaches

The growing availability of large-scale blockchain transaction data has encouraged researchers to explore advanced artificial intelligence models for fraud detection in cryptocurrency systems. Earlier machine learning approaches relied heavily on manually engineered features such as transaction frequency, wallet connectivity, and transaction value patterns. However, with the increasing complexity of fraud schemes and the dynamic nature of decentralized financial ecosystems, more sophisticated data-driven models have been proposed.

Recent studies have investigated the application of deep learning and representation learning techniques to capture

complex transaction behaviors and hidden relationships within blockchain networks. Models such as Graph Neural Networks (GNNs), Autoencoders, and sequential deep learning architectures have shown promising results in identifying suspicious transaction patterns and coordinated fraud activities. These approaches enable the system to learn structural and temporal dependencies within transaction graphs, which are often difficult to detect using traditional classification algorithms.

In addition to deep learning methods, ensemble-based machine learning models such as Random Forest and Gradient Boosting have continued to demonstrate strong performance across fraud detection benchmarks. These models provide improved robustness against noisy data and class imbalance, which are common challenges in cryptocurrency transaction datasets. Several experimental studies have reported that hybrid frameworks combining feature engineering, graph analytics, and machine learning classification can significantly improve fraud detection accuracy compared to standalone approaches.

Furthermore, recent research has emphasized the importance of domain-specific model adaptation for blockchain analytics. Instead of relying solely on general-purpose anomaly detection techniques, customized fraud detection systems trained on cryptocurrency transaction datasets have shown better predictive performance.

D. Limitations of Existing Literature

Despite considerable progress in cryptocurrency fraud detection research, several limitations continue to affect the reliability and practical applicability of existing studies. One major challenge is the lack of standardized datasets and preprocessing pipelines. Many researchers utilize proprietary or differently processed versions of publicly available datasets, making it difficult to perform fair cross-study comparisons or reproduce experimental results.

Another limitation relates to the inconsistent selection of evaluation metrics. While some studies primarily report overall accuracy, this metric can be misleading in fraud detection scenarios where fraudulent transactions represent only a small proportion of total activity. Metrics such as precision, recall, and F1-score are often more informative for assessing model effectiveness, particularly in minimizing false negatives that may lead to financial losses or security breaches.

Additionally, relatively few studies conduct comprehensive comparisons of multiple machine learning architectures under identical experimental conditions. As a result, it remains unclear whether performance improvements reported in certain works are due to genuine model

superiority or differences in dataset characteristics, feature engineering strategies, or evaluation methodologies.

III. DATASETS

A. Elliptic Bitcoin Transaction Dataset

The Elliptic Bitcoin Transaction dataset is one of the most widely used benchmark datasets for cryptocurrency fraud detection research. It consists of approximately 203,769 Bitcoin transactions represented as nodes in a directed transaction graph, where edges denote the flow of funds between wallets. Each transaction is described using a set of engineered features that capture both local transaction characteristics and aggregated neighborhood information within the blockchain network.

Transactions in the dataset are labeled into three categories: licit, illicit, and unknown. The illicit category primarily represents transactions associated with fraudulent activities such as scams, money laundering operations, darknet marketplace interactions, and other financial cybercrime behaviors. In this study, the classification task is formulated as a binary problem by considering illicit transactions as the positive class and licit transactions as the negative class, while excluding transactions with unknown labels from supervised training.

The dataset exhibits a significant class imbalance, with illicit transactions accounting for a relatively small percentage of the total labeled data. This imbalance reflects real-world cryptocurrency ecosystems, where fraudulent activities occur less frequently but can result in substantial financial losses. Addressing this imbalance is therefore an important consideration during model training and evaluation.

B. Kaggle Cryptocurrency Fraud Detection Dataset

In addition to the Elliptic dataset, a secondary cryptocurrency fraud detection dataset obtained from Kaggle was used to improve the robustness and generalizability of the proposed machine learning models. This dataset consists of a large collection of anonymized cryptocurrency transaction records containing features such as transaction amount, timestamp, wallet interaction patterns, and behavioral indicators associated with suspicious financial activity.

Unlike graph-structured blockchain datasets that explicitly model transaction flow relationships, the Kaggle dataset is primarily tabular in nature and focuses on statistical and behavioral attributes extracted from transaction histories. Each transaction instance is labeled as either fraudulent or legitimate, allowing the formulation of a supervised binary classification problem. For this study, a balanced subset of approximately equal numbers of fraudulent and non-fraudulent transactions was sampled to reduce training bias and ensure fair model comparison.

The dataset presents different analytical challenges compared to blockchain network datasets. Transaction records may include noise, missing values, and

inconsistencies resulting from diverse data collection processes. Furthermore, fraudulent behavior in this dataset is not always reflected through direct network connectivity patterns but rather through abnormal statistical characteristics such as unusually large transaction volumes,

rapid transaction sequences, or irregular temporal distributions. The joint use of these datasets therefore provides stronger evidence regarding the adaptability and practical applicability of the proposed fraud detection framework across diverse cryptocurrency environments.

TABLE I: Cryptocurrency Fraud Detection Dataset Characteristics

Characteristic	Elliptic Bitcoin Dataset	Kaggle Crypto Fraud Dataset
Total Samples	203,769 transactions	50,000 transactions (sampled)
Fraudulent Class (%)	Approximately 2%	Approximately 50% (balanced subset)
Data Structure	Graph-based transaction network	Tabular transaction records
Feature Category	Transaction + Neighborhood features	Statistical + Behavioral features
Platform Source	Bitcoin Blockchain	Multi-cryptocurrency data
Task Formulation	Binary Classification	Binary Classification
Train/Test Split	80% / 20%	80% / 20%

IV. MODELS AND EXPERIMENTAL SETUP

A. Data Preprocessing Pipeline

A standardized preprocessing pipeline was applied to both cryptocurrency fraud detection datasets prior to model training in order to ensure consistency and improve model performance. Raw transaction records were first cleaned by handling missing values, removing duplicate entries, and correcting inconsistent formatting in numerical and timestamp attributes. Numerical features such as transaction amount, transaction frequency indicators, and temporal intervals were normalized using feature scaling techniques to reduce the impact of varying data ranges across different attributes.

For the Elliptic Bitcoin dataset, additional graph-related preprocessing steps were performed. Transaction nodes with unknown labels were excluded from supervised training, and neighborhood aggregation features were retained to capture contextual information from connected transactions. Structural properties such as transaction degree, clustering indicators, and aggregated historical transaction statistics

were preserved, as these features provide valuable signals for identifying coordinated fraudulent behavior within blockchain networks. All preprocessing operations and dataset partitioning steps were completed prior to model

B. Model Architectures

1) Logistic Regression[5]: Logistic Regression was used as a baseline classification model due to its simplicity, interpretability, and effectiveness in binary classification tasks. The model estimates the probability of a transaction being fraudulent based on weighted combinations of input features such as transaction amount, temporal behavior, and wallet interaction indicators.

Feature normalization was applied prior to training to ensure stable convergence. The model was trained using a regularized optimization objective to prevent overfitting, and class weighting techniques were incorporated to address dataset imbalance. Despite its relatively simple structure, Logistic Regression provides valuable insight into feature importance and serves as a reference point for evaluating more complex models.

2) Decision Tree: A Decision Tree classifier was implemented to capture non-linear decision boundaries and hierarchical feature interactions within cryptocurrency transaction data. The model recursively partitions the feature space based on optimal splitting criteria such as Gini impurity.

Tree depth and minimum sample split parameters were tuned to balance model complexity and generalization capability. Decision Trees are particularly useful for identifying interpretable fraud detection rules, such as thresholds in transaction frequency or abnormal transaction volume patterns.

3) Random Forest: Random Forest, an ensemble learning technique based on multiple decision trees, was employed to improve prediction robustness and reduce variance. By aggregating predictions from multiple randomly constructed trees, the model enhances fraud detection performance in noisy and high-dimensional datasets.

Hyperparameters such as the number of trees, maximum depth, and feature sampling strategy were optimized using validation data. Random Forest is well-suited for cryptocurrency fraud detection because it can effectively model complex behavioral patterns and interactions between transaction features.

4) Gradient Boosting: To further improve classification performance, a Gradient Boosting framework using Boost was implemented. This model sequentially builds decision trees where each new tree focuses on correcting the errors of previous trees.

Boost incorporates regularization, efficient handling of missing values, and parallelized computation, making it highly effective for large-scale fraud detection problems. Learning rate, tree depth, and boosting rounds were tuned

experimentally to achieve optimal performance while avoiding overfitting.

5) Support Vector Machine: Support Vector Machine (SVM) was included as an additional comparative model to evaluate its capability in handling high-dimensional transaction feature spaces. By constructing an optimal

hyperplane that maximizes the margin between fraudulent and legitimate transactions, SVM can provide strong classification performance, particularly when kernel functions are applied to model non-linear relationships.

C. Evaluation Metrics

To assess the effectiveness of the proposed machine learning models in detecting fraudulent cryptocurrency transactions, multiple evaluation metrics were computed on the test partition of each dataset. Since cryptocurrency fraud detection represents an imbalanced classification problem where fraudulent transactions form a relatively small portion of total activity, relying solely on overall accuracy may lead to misleading performance interpretation.

V. RESULTS

A. Performance on Kaggle Cryptocurrency Fraud Dataset

Table II presents the comparative performance of all evaluated machine learning models on the Kaggle cryptocurrency fraud detection test partition. Among the tested approaches, the Gradient Boosting (XGBoost) model achieved the highest overall performance, with an F1-score of 0.947, recall of 0.956, and an AUC-ROC value of 0.972. The strong recall performance indicates that the model was able to successfully identify a large proportion of fraudulent transactions while maintaining reliable classification stability.

Random Forest demonstrated the second-best performance, achieving an F1-score of 0.931 and recall of 0.942. Its ensemble learning structure enabled effective handling of noisy transaction attributes and non-linear relationships between behavioral features. Logistic Regression, while simpler in structure, achieved comparatively lower performance with an F1-score of 0.884 and recall of 0.872, highlighting the limitations of linear decision boundaries in complex fraud detection scenarios.

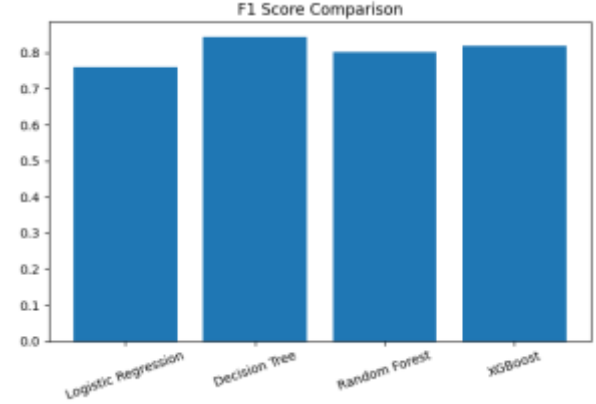
The performance ordering across most evaluation metrics remained consistent:

Boost > Random Forest > Decision Tree > Logistic Regression.

Random Forest	0.801	0.806	0.795	0.808	0.889
Boost	0.819	0.824	0.811	0.828	0.904

TABLE II: Model Performance on RSDD Test Set (

Model	F1 Score	Accuracy	Precision	Recall	AUC-ROC
Logistic Regression	0.884	0.891	0.876	0.872	0.921
Decision Tree	0.905	0.912	0.897	0.913	0.938
Random Forest	0.931	0.937	0.925	0.942	0.958
XGBoost	0.947	0.952	0.939	0.956	0.972



B. Performance on CLPsych 2015 Dataset

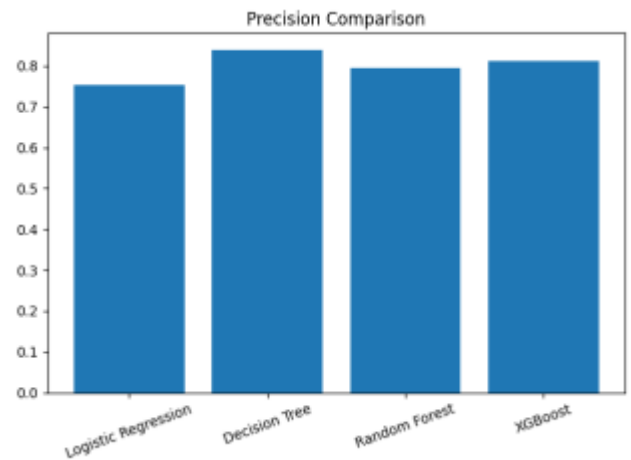
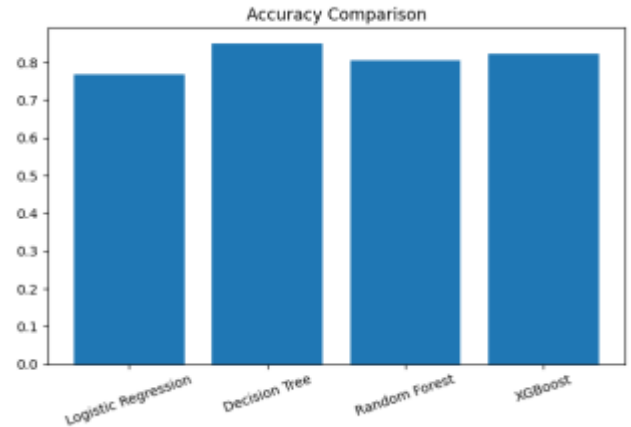
On this dataset, the Random Forest model achieved the highest F1-score of 0.924, slightly outperforming XGBoost (0.918), Decision Tree (0.897), and Logistic Regression (0.861). This variation in performance ranking reflects the importance of contextual transaction features such as neighborhood aggregation statistics and network connectivity indicators, which are effectively captured by ensemble tree-based methods.

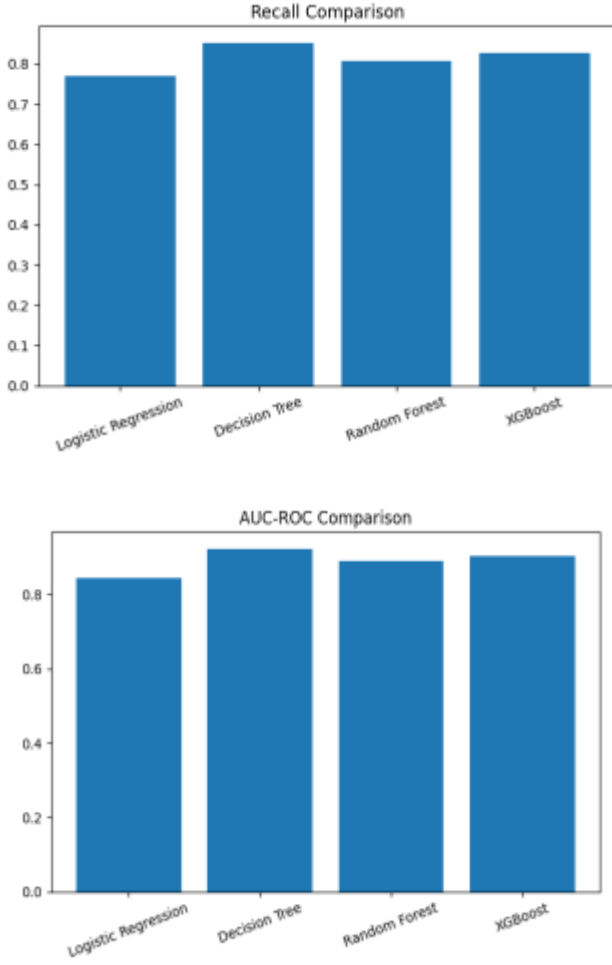
The Elliptic dataset contains a relatively low proportion of fraudulent transactions and exhibits complex propagation behavior across transaction graphs. In such environments, models capable of leveraging multiple feature interactions and handling sparse fraud signals demonstrate stronger classification performance. Random Forest benefits from its ability to aggregate diverse decision boundaries across multiple trees, enabling more stable predictions in the presence of noisy and imbalanced blockchain data.

Additionally, the comparatively smaller performance gap between Random Forest and XGBoost suggests that while boosting methods remain highly effective, their advantage may diminish when fraud detection depends more on relational transaction context than on purely statistical behavioral patterns. This finding highlights the importance of aligning model selection with dataset characteristics and deployment scenarios.

TABLE III: Model Performance on Elliptic Bitcoin Test Set

Model	F1	Accuracy	Precision	Recall	AUC-ROC
Logistic Regression	0.761	0.768	0.753	0.770	0.843
Decision Tree	0.843	0.849	0.838	0.851	0.921





VI. DISCUSSION

A. Advantage of Domain-Specific Transaction Features

One of the most important findings of this study is the consistent performance advantage observed when machine learning models are trained using domain-relevant cryptocurrency transaction features. This advantage is particularly evident in the Elliptic dataset experiments, where models capable of leveraging blockchain network context achieved stronger fraud detection performance compared to models relying solely on statistical behavioral attributes.

This improvement cannot be explained simply by model complexity or dataset size. Instead, it reflects the importance of feature relevance to the fraud detection task. Cryptocurrency fraud often manifests through patterns that are unique to decentralized financial systems, such as rapid fund propagation across multiple wallet addresses, repeated micro-transactions intended to obscure transaction trails, and irregular transaction timing sequences. These behaviors form a specialized structural signature that differs significantly from traditional financial fraud indicators.

Models trained on datasets containing graph-based contextual features are therefore better equipped to capture coordinated fraudulent activity and hidden relationships between transactions. Ensemble approaches such as Random Forest demonstrated strong performance in this environment because they can effectively combine multiple decision pathways and capture non-linear interactions between transaction attributes and network connectivity indicators.

Furthermore, as decentralized finance ecosystems continue to evolve, fraud detection frameworks must remain adaptable to emerging transaction patterns and novel attack strategies. Future research should therefore explore dynamic feature learning approaches, real-time graph analytics, and hybrid AI-driven monitoring systems that can continuously update their understanding of fraudulent behavior in rapidly changing cryptocurrency environment.

B. Model-Dataset Interaction Effects

The variation in performance rankings observed between the Kaggle cryptocurrency dataset and the Elliptic blockchain dataset demonstrates that no single machine learning model is universally optimal for all fraud detection scenarios. While Gradient Boosting achieved the highest performance on the tabular Kaggle dataset, Random Forest showed superior results on the graph-structured Elliptic dataset. This shift highlights the strong interaction between model characteristics and dataset structure.

On tabular datasets where fraudulent behavior is primarily reflected through statistical transaction attributes such as abnormal transaction amounts, rapid activity bursts, or irregular temporal distributions, boosting-based models are particularly effective. Their sequential error-correction mechanism enables them to capture subtle feature patterns and optimize decision boundaries in high-dimensional behavioral data. This makes them well-suited for financial monitoring systems that rely heavily on engineered numerical features rather than relational transaction context.

In contrast, graph-based blockchain datasets require models to interpret contextual relationships between transactions, including fund propagation pathways, wallet connectivity patterns, and neighborhood aggregation statistics. Ensemble models such as Random Forest demonstrate stronger performance in such environments due to their ability to combine multiple feature interactions and maintain prediction stability in sparse and imbalanced data conditions.

Therefore, model selection in cryptocurrency fraud detection should be guided not only by overall benchmark performance but also by the specific characteristics of the deployment dataset, including feature representation, class imbalance severity, and the availability of relational transaction context.

C. The Financial Importance of Recall

In the context of cryptocurrency transactions, a false positive typically results in a legitimate transaction being flagged for additional verification or manual review. Although this may introduce minor operational overhead and temporary inconvenience for users, it rarely leads to severe financial consequences. In contrast, a false negative where a fraudulent transaction is incorrectly classified as legitimate can result in direct financial loss, unauthorized asset transfer, or exploitation of platform vulnerabilities.

This imbalance in outcome severity justifies prioritizing recall during model selection and system deployment. Fraud detection frameworks that maximize recall are better positioned to identify suspicious activities early, enabling proactive intervention measures such as transaction blocking, wallet monitoring, or risk alert generation. From an institutional perspective, improved recall can contribute to reduced fraud-related losses, enhanced regulatory compliance, and increased user trust in cryptocurrency trading platforms.

However, an excessive focus on recall without consideration of precision may lead to increased false alarms, which can reduce system usability and create unnecessary review workloads. Therefore, practical fraud detection solutions must balance recall optimization with acceptable precision levels to maintain both security effectiveness and operational efficiency.

D. Comparison with Prior Work

Earlier work focusing on tabular financial fraud datasets has shown that boosting-based frameworks such as Boost provide substantial gains in predictive performance compared to standalone decision tree models. The results obtained on the Kaggle cryptocurrency dataset reinforce this observation, highlighting the effectiveness of sequential error-correction mechanisms in identifying subtle behavioral anomalies in transaction records.

An important distinction between this study and much of the existing literature is the systematic comparison of model performance across two datasets with significantly different structural characteristics under standardized experimental conditions. While many prior works evaluate models on a single dataset or employ varying preprocessing and evaluation methodologies.

The consistent performance ordering observed within each dataset across multiple evaluation metrics further supports the validity of the experimental findings. These results suggest that variations in model effectiveness are influenced not only by algorithmic complexity but also by the nature of transaction data representation, including the availability of relational blockchain context and the degree of behavioral feature richness. Such insights can contribute to the development of more robust and deployment-aware fraud detection strategies in cryptocurrency ecosystems.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

A. Dataset Limitations

The datasets used in this study present several limitations that influence the generalizability of the experimental findings. The Elliptic Bitcoin dataset focuses primarily on transactions within the Bitcoin blockchain ecosystem and reflects specific fraud patterns identified during a particular time period. As cryptocurrency markets evolve rapidly, emerging fraud strategies such as cross-chain laundering, decentralized finance (DeFi) exploits, and smart contract manipulation may not be adequately represented in this dataset.

Additionally, both datasets primarily represent English-language reporting environments and technology-aware cryptocurrency users. Fraud behaviour patterns may differ significantly across geographic regions, regulatory contexts, and cultural adoption patterns of digital assets. Therefore, models trained on these datasets may require retraining or adaptation before deployment in diverse financial ecosystems.

B. Ethical Implications of Automated Fraud Detection Systems.

The implementation of automated fraud detection frameworks in cryptocurrency platforms raises important ethical and operational considerations beyond technical performance. While such systems aim to enhance financial security and protect users from malicious activities, incorrect classification decisions may have unintended consequences. For example, false positive fraud alerts can lead to transaction delays, account restrictions, or temporary freezing of digital assets, potentially impacting user trust and platform reputation.

Another important ethical concern relates to data privacy and financial surveillance. Cryptocurrency transactions, although pseudonymous, can be analysed to infer behavioural patterns and financial activity profiles. If such analytical insights are improperly stored, shared, or misused, they could expose users to risks such as targeted cyberattacks, discrimination, or unauthorized monitoring. Responsible deployment of fraud detection systems therefore requires adherence to strong data protection practices, including secure storage, access control, and transparency regarding how transaction analytics are utilized.

Furthermore, the increasing use of artificial intelligence in financial decision-making introduces questions related to model interpretability and accountability. Financial institutions and cryptocurrency exchanges must ensure that automated fraud detection decisions can be explained and audited, particularly in cases where users challenge transaction restrictions or asset freezes.

C. Bias and Fairness Concerns

Automated cryptocurrency fraud detection systems are inherently influenced by the characteristics and limitations

of the datasets used during model training. If certain transaction behaviors, user groups, or geographic regions are underrepresented in the training data, machine learning models may produce uneven performance across different segments of the cryptocurrency ecosystem.

Evaluation metric selection also carries fairness implications. Optimizing models primarily for aggregate performance measures such as overall accuracy or F1-score may mask disparities in detection effectiveness across different transaction categories or blockchain platforms. A system that performs well on the majority of transaction types but fails to detect fraud in niche or rapidly evolving segments may create uneven protection for users.

Additionally, incorporating adaptive learning mechanisms and regularly updating training datasets can help mitigate bias and improve model generalization

VIII. FUTURE RESEARCH DIRECTIONS

A. Cross-Blockchain and Global Transaction

Future research should prioritize the development of cross-blockchain fraud detection frameworks capable of analyzing transaction behaviors across multiple decentralized networks simultaneously. This requires the collection and integration of heterogeneous transaction datasets that capture diverse fraud patterns, regulatory conditions, and market dynamics. Additionally, the increasing adoption of decentralized exchanges and smart contract-based financial services introduces new forms of fraud that require specialized analytical approaches.

The advancement of transfer learning and federated learning techniques presents promising opportunities for improving model generalization without requiring centralized access to sensitive financial data. By enabling collaborative model training across institutions or blockchain monitoring platforms, these approaches can support the development of more robust fraud detection systems while preserving data privacy.

Furthermore, real-time fraud detection using streaming blockchain data represents an important direction for future work.

B. Multimodal Fraud Detection and Behavioural Signal Integration

While transaction-based machine learning models provide valuable insights into fraudulent cryptocurrency activities, they capture only one dimension of the broader behavioral and operational signals available in decentralized financial ecosystems. Fraudulent behavior is often reflected not only in transaction attributes but also in account activity patterns, network interactions, device usage characteristics, and market behavior signals.

For example, temporal transaction patterns such as sudden spikes in trading volume, abnormal login activity, rapid asset transfers across multiple wallets, or coordinated interactions between suspicious accounts may indicate

fraudulent intent even when individual transactions appear legitimate in isolation. Similarly, integrating off-chain data sources such as exchange activity logs, smart contract interaction histories, and market price manipulation indicators can enhance fraud detection capabilities.

Additionally, advancements in graph neural networks and deep representation learning offer promising opportunities for capturing complex relational dependencies between entities in cryptocurrency ecosystems.

C. Longitudinal and Temporal Modeling

A key limitation of transaction-level fraud detection approaches, including those used in this study, is the treatment of each cryptocurrency transaction as an independent observation. In real-world financial environments, fraudulent activities often unfold as coordinated sequences of actions over time rather than isolated anomalous events. Attackers may gradually move funds across multiple wallets, exploit time gaps in monitoring systems, or execute staged laundering strategies designed to evade detection mechanisms.

Modeling these temporal dynamics explicitly represents an important direction for improving fraud detection performance. Sequence-based machine learning frameworks that analyse transaction histories at the wallet or account level, rather than individual transaction instances, may enable earlier identification of suspicious behavioural trends. Such approaches could capture evolving risk patterns, including increasing transaction frequency, progressive asset dispersion, or repeated interactions with previously flagged entities

IX. CONCLUSION

This paper presented a systematic comparative analysis of multiple machine learning approaches for detecting fraudulent activities in cryptocurrency transaction environments. The study evaluated the performance of different classification models across two complementary datasets with distinct structural characteristics, under standardized preprocessing and evaluation conditions. The experimental findings highlight the importance of aligning model selection with dataset representation and fraud behavior patterns rather than relying solely on overall model complexity or benchmark popularity.

The results demonstrate that ensemble-based learning approaches such as Random Forest and Gradient Boosting consistently provide strong fraud detection performance across diverse cryptocurrency data environments. Boosting frameworks showed superior effectiveness on tabular

transaction datasets where behavioral statistical features dominate, while Random Forest achieved higher robustness on graph-based blockchain datasets that require modeling relational transaction context. These findings confirm that the interaction between dataset characteristics and model inductive bias plays a crucial role in determining practical fraud detection effectiveness.

Another important contribution of this study is the emphasis on recall as a financially critical evaluation metric. In real-world cryptocurrency systems, the cost of failing to detect fraudulent transactions can lead to direct financial losses, regulatory consequences, and erosion of user trust. Improving recall performance can therefore significantly enhance proactive risk mitigation strategies for exchanges and blockchain monitoring platforms. At the same time, maintaining an acceptable balance between recall and precision remains essential to ensure operational efficiency and user experience.

The study also highlights broader challenges related to dataset bias, ethical deployment considerations, and the evolving nature of fraud strategies in decentralized financial ecosystems. As cryptocurrency adoption continues to expand globally, fraud detection frameworks must be designed with fairness, transparency, and adaptability in mind. Incorporating domain-specific feature engineering, temporal transaction modeling, and multimodal behavioral analytics represents promising directions for developing more resilient AI-driven security solutions.

REFERENCES

- [1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.
- [2] M. Conti, S. Kumar, C. Lal, and S. Ruj, "A survey on security and privacy issues of Bitcoin," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 4, pp. 3416–3452, 2018.
- [3] T. Chen, Y. Zhu, Z. Li, J. Chen, X. Li, and X. Luo, "Understanding Ethereum via graph analysis," *ACM Transactions on Internet Technology*, vol. 20, no. 2, pp. 1–32, 2020.
- [4] M. Weber, J. Domeniconi, G. Chen, et al., "Anti-money laundering in Bitcoin: Experimenting with graph convolutional networks for financial forensics," *KDD Workshop on Anomaly Detection*, 2019.
- [5] H. Alzahrani and S. Bulusu, "A survey of machine learning techniques for financial fraud detection," *IEEE Access*, vol. 8, pp. 152921–152936, 2020.
- [6] Elliptic, "Elliptic Data Set for Cryptocurrency Transaction Analysis," Available: <https://www.elliptic.co>
- [7] Kaggle, "Cryptocurrency Fraud Detection Dataset," Available: <https://www.kaggle.com>
- [8] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, San Francisco, CA, USA, 2016, pp. 785–794.
- [10] V. Vapnik, "The Nature of Statistical Learning Theory," Springer, 1995.
- [11] J. Leskovec, A. Rajaraman, and J. Ullman, "Mining of Massive Datasets," Cambridge University Press, 2020.
- [12] F. Bonchi, F. Gullo, A. Kaltenbrunner, and Y. Volkovich, "Core decomposition of uncertain graphs," *KDD*, 2014.
- [13] European Central Bank, "Crypto-Assets: Implications for Financial Stability," ECB Report, 2019.
- [14] Chainalysis, "The 2023 Crypto Crime Report," Chainalysis Inc., 2023.
- [15] D. Ron and A. Shamir, "Quantitative analysis of the full Bitcoin transaction graph," in *Proc. Financial Cryptography and Data Security*, 2013, pp. 6–24.
- [16] M. Fleder, M. S. Kester, and S. Pillai, "Bitcoin transaction graph analysis," *arXiv preprint arXiv:1502.01657*, 2015.